

Predicting the long-term impact of AI assistants on collective intelligence

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Abstract

Recent advances in artificial intelligence (AI) have enabled widespread use of AI assistants for various cognitive tasks, ranging from everyday tasks to scientific discovery and creative writing. While AI assistance can improve the performance of each individual, it can weaken their subsequent performance without AI and further limit the diversity of interests and opinions. These risks raise the question of whether human collective intelligence can persist as our society becomes increasingly reliant on AI assistants. To address this question, we develop an evolutionary model of collective intelligence, where each individual participate in a prediction task, consult AI, and imitate each other's prediction strategy. We find that "Chatbot" AI, which depends on users' interests, can prevent the emergence of collective intelligence by either limiting diversity or propagating systematic bias through overreliance. Directly incentivizing or penalizing AI use generally worsens these outcomes. Instead, a balanced incentive structure that preserve minority expertise and reward contributions to collective outcome can promote the emergence of collective intelligence. However, the inherent ability of the population's to make prediction in the absence of AI remains suboptimal even under this balanced incentive structure, raising concerns for continued AI reliance.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability (Galton, 1907; Woolley et al., 2010; Yaniv and Milyavsky, 2007; Hommes et al., 2005; Lorge et al., 1958). This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups (Dreyer et al., 2025; Simons, 2004; Berdahl et al., 2013) and human societies (Woolley et al., 2010; Riedl et al., 2021) to machine-learning systems (Breiman, 1996, 2001). Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences (Mann and Helbing, 2017), evolutionary biology (Wang et al., 2025), and computer science (Werfel, 2026). Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes (Cui and Yasseri, 2024).

The recent emergence of generative AI and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks (Noy and Zhang, 2023; Brynjolfsson et al., 2025), including scientific discovery (Lu et al., 2026) and creative writing (Doshi and Hauser, 2024). In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing (Bick et al., 2026; Sajadieh et al., 2026), the potential long-term impact of AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise (Vaccaro et al., 2024), and opinion aggregation (Burton et al., 2024; Tessler et al., 2024). Conversely, empirical evidence indicates that the use of AI can limit the diversity of opinions (Fügener et al., 2021; Doshi and Hauser, 2024; Lee and Chung, 2024; Meincke et al., 2025), a key requirement for the emergence of CI (Hong and Page, 2004; Lorenz et al., 2011; Aplin et al., 2014; Becker et al., 2017).

Our ability to predict the long-term impact of AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI use. On one hand, many industrial sectors are pushing towards increased AI use, where the number of “tokens” used is often perceived as a proxy for productivity (Bousquette, 2026; Keary, 2026). On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence (Thorp, 2023; Chawla, 2026). Such discrepancy raises a question about whether directly incentivizing or penalizing AI use will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of AI assistants. Such incentive structure should seek to retain the benefits of AI and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual participates in a prediction task from partial observations of under-

69 lying causal factors (Mann and Helbing, 2017; Wang et al., 2025; Hong et al., 2012).
 70 In particular, we allow each individual to independently consult with AI assistants
 71 and revise their predictions accordingly. The resulting model allows us to investi-
 72 gate the robustness of human CI under different assumptions about the AI model’s
 73 ability to make predictions as well as incentive structures that are associated with
 74 the prediction task. In doing so, we assess the impact of directly incentivizing or
 75 penalizing AI use on CI and further suggest a balanced incentive structure that can
 76 maintain CI in the presence of AI.

77 Mathematical model

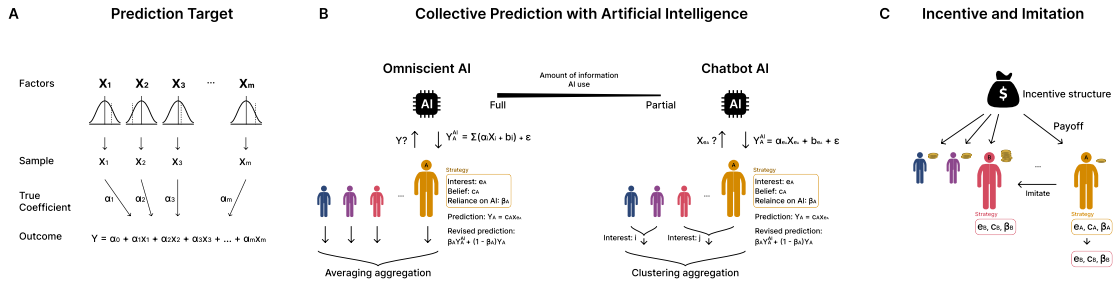


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

78 We begin by extending previously established models of CI, in which a group of
 79 players participates in a prediction task (Fig. 1): each player makes an individual
 80 prediction based on partial observations of the underlying causal factors, and the
 81 group forms a collective prediction by aggregating these individual predictions (Hong
 82 et al., 2012; Mann and Helbing, 2017; Wang et al., 2025). Specifically, the prediction
 83 target, Y , is modeled as a linear combination of M causal factors x_i , with fixed
 84 coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor is drawn from independent
 85 normal distribution (Fig. 1A). Then, each individual A observes a single factor
 86 of their interest x_{e_A} and makes a prediction based on their own belief, c_A : $Y_A =$
 87 $c_A x_{e_A}$ (Fig. 1B). We assume each individual can then independently consult their
 88 AI assistants and revise their prediction based on their reliance on AI, β_A : $Y_A^{re} =$
 89 $\beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Fig. 1B). The revised predictions are then aggregated using two

90 alternative aggregation rules, averaging and clustering, to form collective prediction.
 91 To model AI, we assume that AI has already been trained on similar tasks and
 92 is able to make predictions based on partial or full observations of causal factors
 93 (Fig. 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based
 94 on its internal parameters, which can include factor-specific bias (b_i) and stochastic
 95 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 96 amount of information AI use when making predictions. In one extreme, the AI
 97 can observe all causal factors and use all information to make predictions, regardless
 98 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 99 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 100 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 101 and ‘Chatbot AI’, respectively.

102 We assume that appropriate aggregation rule for collective prediction depends
 103 on the type of AI. When Chatbot AI is used, each revised prediction depends on a
 104 single factor. Thus, we use clustering aggregation, under which revised predictions
 105 are first averaged among individuals who are interested in the same factor, and
 106 then summed across factors. When Omniscient AI is used, however, each revised
 107 prediction contains information about all causal factors, in which case clustering
 108 these predictions by interest would be inappropriate. Instead, we use averaging
 109 aggregation, under which revised individual predictions are averaged directly to form
 110 the collective prediction.

111 Each generation consists of multiple round of predictions, and each player is re-
 112 warded a payoff according to the incentive structure (Fig. 1C). For simplicity, we
 113 assume that a sufficiently large number of prediction rounds occurs within each gen-
 114 eration, allowing us to use the expected payoff value rather than computing the
 115 payoffs for every round. We consider two incentive structures from previous work,
 116 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 117 lective accuracy in the absence of AI (Wang et al., 2025). The feedback structure
 118 gives higher payoff to individuals who can push the collective prediction toward the
 119 true target, regardless of their individual predictive accuracy: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$. In
 120 contrast, the niche-expert structure gives higher payoff to individuals who have both
 121 better predictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$. Here, ρ_{e_A}
 122 denotes the proportion of players interested in the same factor, thereby allowing the
 123 niche-expert structure to preserve interest diversity. We note collective intelligence
 124 fails to emerge under the combination of averaging aggregation with niche-expert
 125 structure in the absence of AI (Wang et al., 2025).

126 Through time, the group evolves by allowing each individual to imitate peers
 127 who received higher payoffs (Fig. 1C). Specifically, after each generation, two indi-
 128 viduals are randomly sampled, and one imitates all three strategies (the factor of
 129 interest, factor belief, and reliance on AI) of the other with a probability determined
 130 by the difference in their expected payoffs (Materials and Methods). Individual
 131 predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation
 132 rules. Collective accuracy is defined as the reduction in prediction error achieved

by the collective prediction relative to a baseline prediction α_0 (Wang et al., 2025):
 $1 - \mathbb{E}[(Y - \hat{Y}^{\text{re}})^2] / \mathbb{E}[(Y - \alpha_0)^2]$. Hereafter, we say that CI emerges when the collective
accuracy reaches a near perfect level. Full derivations of the collective prediction,
expected payoffs, and collective accuracy under the two AI models are provided in
Supplementary Text S1 and S2.

This model now allows us to explore the impact of AI on CI depending on the type
of AI and the incentive structure. Here, we begin by comparing the emergence of CI
with and without AI and subsequently investigate how different incentive structures
affect the emergence of CI in the presence of AI.

AI can either promote or prevent the emergence of CI

To understand the impact of AI, we first compare collective dynamics with and
without AI. First, consistent with Wang et al. (2025), simple imitation rules and
incentive structures allow collective intelligence to emerge in the absence of AI (Fig.
2A,C,D). One exception is the use of simple averaging with niche-expert incentive
structures, under which the collective accuracy of the population remains near zero.
This is because enhancing individual accuracy causes individuals to predict close to
the true coefficient of their interest, but simply averaging these predictions does not
necessarily produce an accurate estimate of the full prediction target (Fig. S1).

We find that AI use can have heterogeneous outcomes depending on its type.
For example, the use of Omniscient AI can either preserve CI under the feedback
incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
expert incentive structure (Fig. 2B). These results are robust across different levels
of AI accuracy (Fig. S2) and can also be characterized analytically in the large-
population limit (Supplementary Text S3.1).

In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
structure, the use of Chatbot AI leads to a loss of diversity in the population's
interest, making it impossible to account for effect of some causal factors (Fig. S2,
S3). This causes an increase in collective variance and stochastic decrease in collective
accuracy (Fig. 2C; Fig. S3). We note that the loss of interest diversity may arise
even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot
AI makes the group more vulnerable to such loss (Fig. S4). Under the niche-
expert incentive structure, the population becomes overly reliant on AI, causing the
collective prediction to inherit the AI's bias (Fig. S3). This leads to an increase in
collective bias and a decrease in collective accuracy (Fig. 2; Fig. S3). Interestingly,
we find that the niche-expert structure can prevent the use of AI under specific
assumptions about the relationship between the true coefficient and AI bias (Fig.
S2). In this case, the population reduces the reliance on AI assistance, allowing
the CI to emerge eventually. However, this phenomenon can substantially delay the

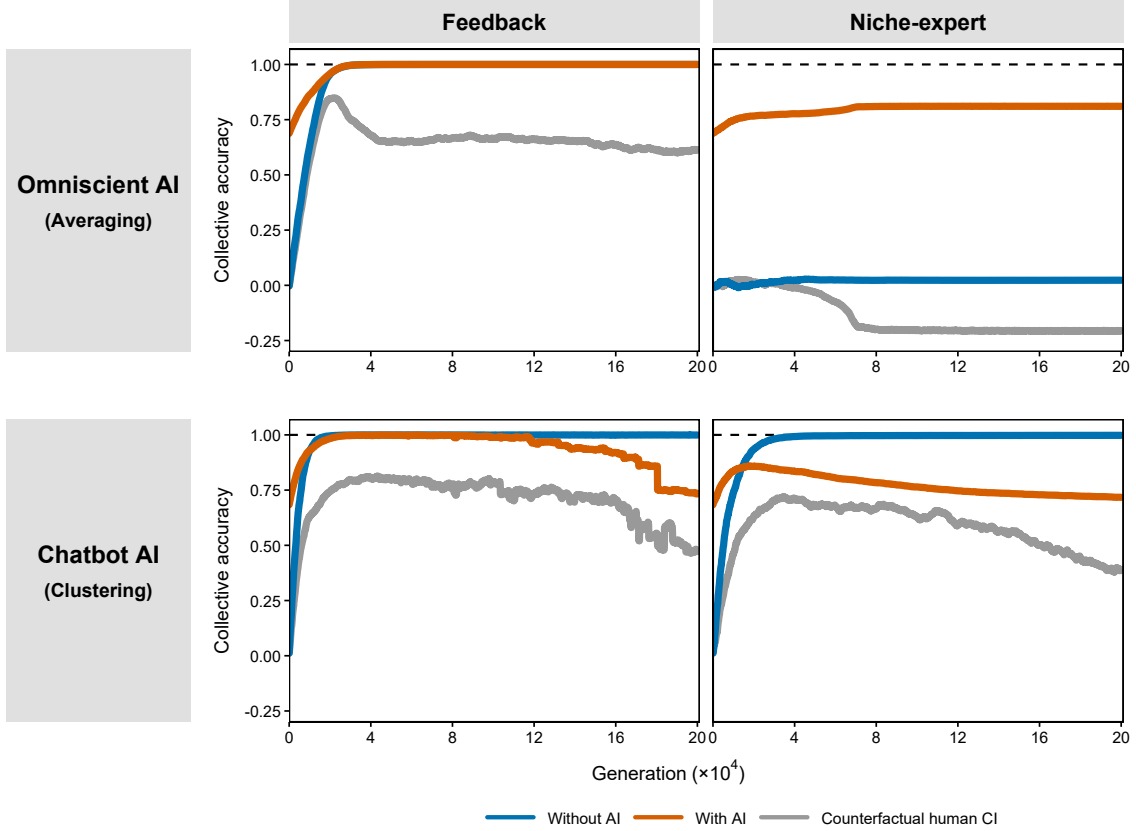


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Blue lines represent trajectories of collective accuracy without AI assistance. Orange lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

173 emergence of CI (Fig. S5).

174 To investigate whether the group can retain its predictive ability when AI assis-
 175 tance suddenly becomes unavailable at any given time, we quantify counterfactual
 176 collective accuracy based on unrevised predictions. Across all scenarios, we find that
 177 the counterfactual collective accuracy is lower than the collective accuracy of the
 178 corresponding population (Fig. 2).

179 In summary, an Omniscient AI can promote CI, but AI operating with limited

180 information may prevent the emergence of CI. Moreover, reliance on AI can limit
 181 inherent ability of the population to make accurate predictions on their own, raising
 182 concerns about future AI use. These findings further underscore the need for incen-
 183 tive structures that can mitigate the potential risks associated with AI assistance.

184 Directly incentivizing or penalizing AI use can ad- 185 ditionally harm CI

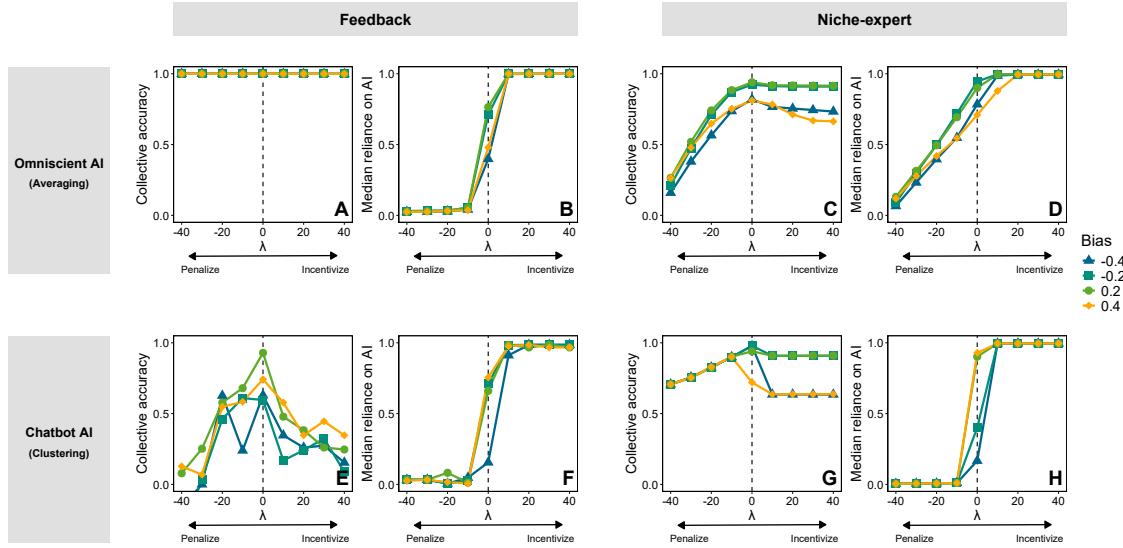


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (Omniscient AI, A–D and Chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

186 Given the strong influence of AI on collective outcomes, a straightforward inter-
 187 vention is to directly incentivize or penalize the use of AI. To investigate whether
 188 such direct incentives or penalties can help the emergence of CI, we modify the two
 189 incentive structures by adding a payoff term proportional to individual reliance on

190 AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

191 where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$), and β_A
192 represent individual reliance on AI (Supplementary Text S1.3 and S2.3).

193 Directly incentivizing or penalizing AI use can further reduce the collective ac-
194 curacy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback
195 structure; in the latter case, the population retains collective intelligence regardless
196 of the intervention strength (Fig. 3A). The reduction in collective accuracy arises be-
197 cause incentivizing or penalizing AI use causes individuals to imitate others based on
198 their AI use rather than their predictive accuracy. Thus, stronger interventions can
199 push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B, D, F, H) but
200 these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When
201 AI bias is relatively large, a weak penalty can lead to higher collective accuracy,
202 but this represents a narrow parameter regime (Figure 3G). Overall, these results
203 indicate that directly rewarding or penalizing AI use, independently of predictive
204 performance, may exacerbate rather than mitigate the risk of AI on CI.

205 **Balanced incentives mitigate AI-induced failures of** 206 **CI**

207 Neither the original incentive structures nor simple interventions targeting AI use are
208 sufficient to promote CI. What incentive structure, then, would work in the presence
209 of AI? Here, we propose a “balanced” incentive structure, modeled as the average of
210 the feedback and niche expert incentives (Supplementary Text S2.3).

211 The core idea behind the balanced structure is to maintain interest diversity and
212 prevent collective bias. On one hand, the feedback incentive structure is prone to
213 the loss of interest diversity but can prevent collective bias by incentivizing players
214 whose predictions lie opposite to the collective prediction. On the other hand, the
215 niche-expert structure can maintain interest diversity because it directly incentivizes
216 minority interests. However the niche-expert structure cannot prevent collective bias
217 because, within the same factor of interest, individuals are rewarded only according
218 to their individual predictive performance, which in turn homogenize the predictions.

219 Therefore, when the two incentive structures are combined, the population achieves
220 CI by maintaining interest diversity while limiting collective bias (Fig. 4A, S6). Un-
221 der the balanced structure, individuals remain more evenly distributed across causal
222 factors, and their beliefs lie closer to the true coefficients than under either incentive
223 structure alone (Fig. 4C–E). The balanced structure also produces higher counter-
224 factual collective accuracy than the feedback or niche-expert structure (Fig. 4B,
225 S6). These patterns are robust to how the two incentive structures are weighted
226 when combined (Fig. S7), and are driven by the emergent property of the balanced
227 structure, under which the population increases its reliance on AI as AI becomes
228 more accurate (Fig S8).

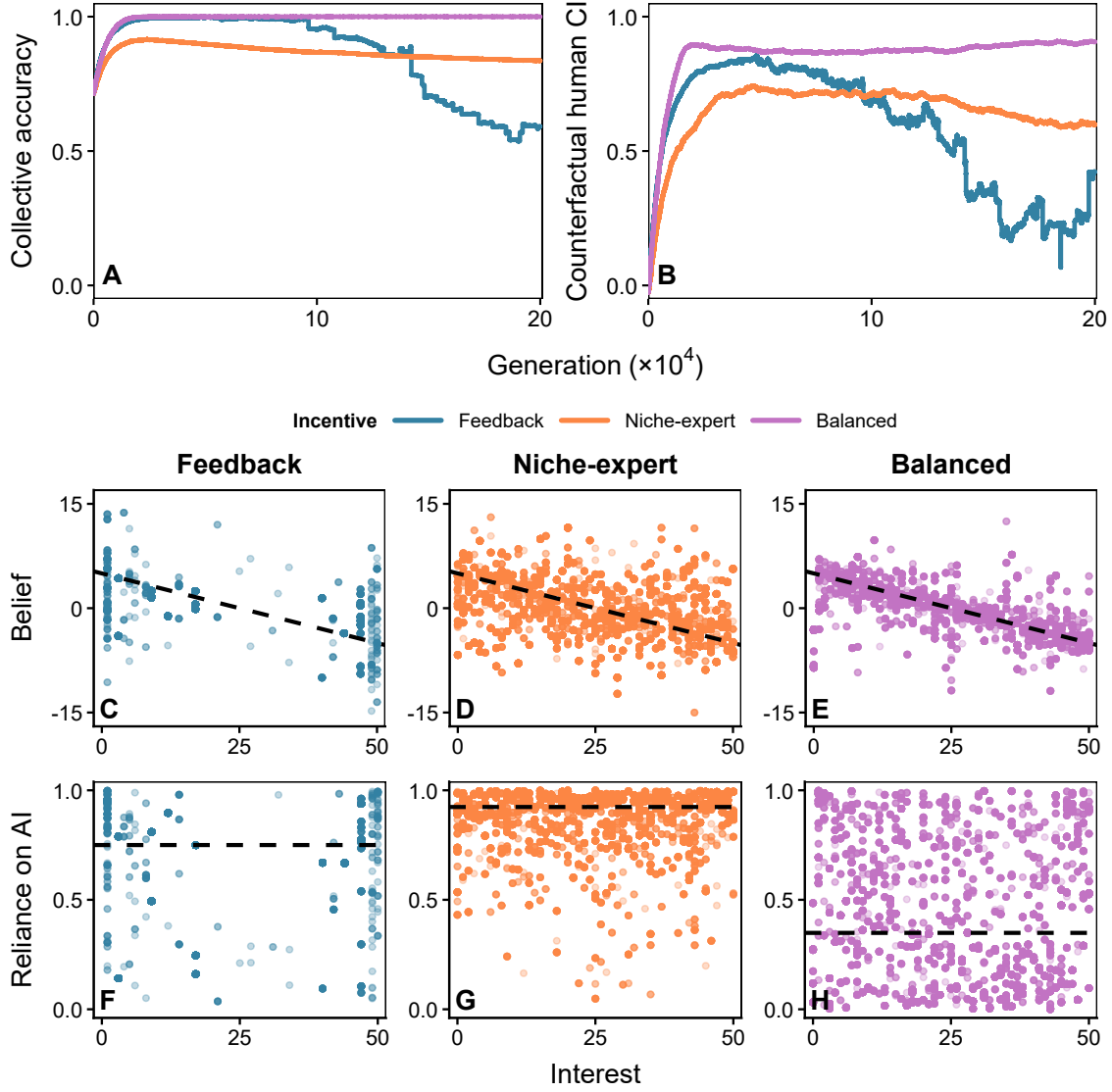


Figure 4: **Balanced structure can promote collective intelligence.** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

Discussion

The ongoing race towards the development of “better” AI continues to pose questions about the societal benefits and risks of AI use (Bender et al., 2021; Burton

et al., 2024; Cui and Yasseri, 2024). Here, we used mathematical models of evolutionary dynamics to investigate how AI use and incentive structures jointly affect the emergence of CI, which is defined as the population’s ability to make accurate predictions. We find that an “Omniscient” AI model, which incorporates information about all relevant causal factors, can improve collective accuracy, whereas a “Chatbot” AI model, whose outcome depends on the user’s interest, can impair it. These contrasting outcomes arise because AI assistance affects not only individual predictions, but also the population’s interest diversity, beliefs, and reliance on AI. We also show that directly incentivizing or penalizing AI use, independent of individual’s performances, can further limit the collective accuracy of the population. Instead, a balanced incentive structure can maintain CI by promoting diversity and limiting bias.

Our analysis reveals that AI can prevent the emergence of CI through two different mechanisms. First, AI use can cause individuals to concentrate on a subset of the factors that determine the prediction target, thereby reducing diversity in their interests. This result is broadly consistent with recent empirical studies showing that AI assistance can improve individual performance but limit the diversity of outputs, in domains ranging from scientific research (Hao et al., 2026) to creative writing (Doshi and Hauser, 2024). Second, over-reliance on AI can cause the population to inherit inherent biases in AI’s predictions (Fügener et al., 2021; Vicente and Matute, 2023), causing the prediction accuracy to decrease eventually. This mechanism can prevent the emergence of CI even when the interest diversity is maintained because biases in AI predictions can be amplified when individual predictions are aggregated. Thus, preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI use need to be calibrated to prevent systematic AI biases from becoming collective biases (Buçinca et al., 2021).

Our analysis of counterfactual collective accuracy also raises an important concern for future AI use: while AI use may improve the apparent performance of individuals and, in some cases, the population as a whole, it can simultaneously reduce the population’s ability to make accurate predictions without AI. These results echo findings from a recent field experiment, which demonstrated that students, who seemingly performed better in solving mathematics problems under AI assistance, performed worse on their own than those who did not have access (Bastani et al., 2025). These findings highlight the need to evaluate AI tools not only by the performance gains they provide during use, but also by whether individuals and populations retain the ability to perform without them (Melumad and Yun, 2025; Ke et al., 2026). Making this distinction clear will be particularly important for education, where AI may help students complete a task without helping them learn how to do it on their own (Bastani et al., 2025; Melumad and Yun, 2025).

As generative AI and large language models are already generating substantial benefits, many organizations, both industrial and academic, are increasingly seeking to improve their collective performance by regulating AI use (Thorp, 2023; Bousquette, 2026; Chawla, 2026; Keary, 2026). Such regulations may take direct forms,

including AI-use policies or token-based leaderboards, or may operate indirectly through other incentives, such as productivity rewards or the social stigma associated with AI use (Reif et al., 2025). Our model suggests that directly incentivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups to propagate AI biases, whereas penalizing AI use could lead to underreliance, preventing groups from using AI even when it performs well in the relevant domain. Our result instead shows that appropriate AI use and accurate CI can be promoted by focusing on the traditional properties of groups that have been emphasized for decades: maintaining diversity and preventing collective bias, rather than on AI use itself (Hong and Page, 2004; Lorenz et al., 2011; Mann and Helbing, 2017).

There are several limitations to our analyses. First, our model allows for social learning among humans, but AI’s inherent ability remains fixed throughout the evolutionary process. In reality, AI can also learn from human-generated data (Ouyang et al., 2022). Such bidirectional learning could increase the similarity between human and AI errors and thereby reduce group performance (Steyvers et al., 2022). Second, we model AI errors using fixed, factor-specific additive biases and independent random errors. This assumption allows random errors to be reduced through aggregation, while systematic biases can still persist in the collective prediction. In reality, AI errors may have more complex structures, including biased estimates of factor effects and correlated errors across individuals, factors, or repeated interactions. Further empirical and theoretical work is therefore needed to understand how these error structures affect the emergence of CI. Finally, following established models of CI, we assume that the prediction target is a linear combination of independent causal factors. This assumption does not hold for many real-world tasks, where causal factors may be correlated or interact nonlinearly. Future studies should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

The rapid adoption of AI assistants is changing how individuals learn, work, and make decisions (Tessler et al., 2024; Glickman and Sharot, 2025; Sajadieh et al., 2026; Bick et al., 2026; Ke et al., 2026). While the immediate benefits and risks of these systems have been evaluated at the individual level (Noy and Zhang, 2023; Bastani et al., 2025), our study raises a broader concern that their widespread AI use may reshape collective intelligence of human society. Our results further underline the importance of designing reward structures that preserve minority expertise and prevent AI biases from persisting across the population (Mann and Helbing, 2017; Vicente and Matute, 2023; Glickman and Sharot, 2025; Wang et al., 2025). A detailed empirical understanding of how AI assistance reshapes diversity, expertise, and reliance at the population level will therefore be critical for predicting the dynamics of future of human collective intelligence.

315 Methods

316 Prediction target

317 Following previous models of CI, we model the prediction target as a linear combi-
318 nation of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017;
319 Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

320 where the coefficient α_i determines the magnitude and direction of the effect of
321 factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an
322 independent Gaussian random variable with mean zero and standard deviation σ_i .

323 Players

324 We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel
325 prediction task. In the absence of AI assistance, the strategy of player A is char-
326 acterized by two variables: interest e_A and belief c_A . Player A observes the causal
327 factor X_{e_A} and makes a prediction by multiplying the observed factor value by their
328 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

329 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does
330 not observe any causal factor and instead makes a constant prediction based solely
331 on their belief. We refer to this as a stubborn strategy because the player’s predic-
332 tion does not respond to the realized values of the causal factors. For notational
333 convenience, we assume that $X_0 = 1$.

334 When the group is assisted by AI, each player’s strategy additionally includes
335 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
336 pendently and revises their prediction by taking a weighted average of their own
337 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

338 where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI,
339 whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

340 AI models

341 We assume that the AI model has been trained on similar tasks and is therefore
342 capable of making predictions based on causal factors. However, because the model
343 is imperfect, its outputs are subject to systematic additive bias and stochastic error.
344 The additive bias associated with factor i is denoted by b_i , and the stochastic error
345 is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its
346 biases therefore remain fixed throughout the simulation. For simplicity, we assume

347 $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows
 348 biases to differ among factors.

349 We assume that the stochastic error follows a Gaussian distribution with mean
 350 zero and standard deviation τ . Each time a player consults the AI, an indepen-
 351 dent error is generated. Errors are therefore independent across players and across
 352 repeated consultations by the same player. Consequently, stochastic AI errors can
 353 be attenuated when independently generated predictions are aggregated across play-
 354 ers. We therefore hold τ constant and vary the magnitude of the additive bias to
 355 manipulate the level of AI accuracy during our analyses.

356 We consider two extreme types of AI that differ in the amount of information
 357 available for prediction. The first is termed the Omniscient AI, which can observe
 358 all M causal factors and incorporate them into its prediction regardless of the user's
 359 interest. The prediction generated by the Omniscient AI for player A is expressed
 360 as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

361 The second is termed the Chatbot AI, which can use only the single causal factor
 362 X_{e_A} associated with the player's interest. The prediction generated by the Chatbot
 363 AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

364 Aggregation rules

365 The revised individual predictions are aggregated to form a collective prediction.
 366 We use two aggregation rules proposed in previous research (Wang et al., 2025),
 367 depending on the type of AI used by the group. First, when the group is assisted by
 368 the Omniscient AI, the AI component of each revised prediction contains information
 369 about all M factors, even though the individual's own prediction is based on only
 370 one factor. Each revised prediction can therefore be treated as an estimate of the
 371 entire target. Accordingly, we calculate the collective prediction by taking the simple
 372 average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (7)$$

373 Second, when the group is assisted by the Chatbot AI, both the player's initial
 374 prediction and the AI's prediction are based on the same causal factor. Each revised
 375 prediction therefore contains information about only one factor and can be treated
 376 as an estimate of that factor's contribution to the target. We consequently use
 377 clustering aggregation.

378 Clustering aggregation is performed by grouping individuals according to their
 379 interests and calculating the mean prediction within each group. The collective
 380 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs based on the underlying incentive structure. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player A 's revised prediction Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

The niche-expert structure assigns higher payoffs to individuals who make accurate predictions while having an underrepresented interest. Specifically, the squared prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

Previous research has shown that these two incentive structures can produce near perfect collective accuracy through social learning, except when the niche-expert incentive is combined with simple averaging (Wang et al., 2025). They are therefore used as baseline structures for examining how AI assistance affects the emergence of CI.

To examine the effects of directly encouraging or discouraging AI use, we modify the feedback and niche-expert structures by adding a term proportional to the player's reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

Here, λ represents the strength of the direct incentive or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to the original feedback and niche-expert structures.

The balanced structure is defined as the arithmetic mean of the feedback and niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

414 Simulations

415 All simulations are conducted using an agent-based model with 50 causal factors
 416 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 417 target are sampled at the beginning of the simulations and subsequently held fixed.

418 Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the
 419 simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization.
 420 Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

421 Players' initial strategies are also sampled from predefined probability distribu-
 422 tions. The distribution of individual belief c_A depends on the aggregation rule. Under
 423 simple averaging, each factor's contribution is scaled by the proportion of players in-
 424 terested in that factor. We therefore sample beliefs from a Gaussian distribution
 425 with mean zero and standard deviation 100, following the scaling used in previous
 426 work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a
 427 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 428 are averaged separately within each interest group and then summed, no additional
 429 scaling of c_A is required. The initial interest of each player is sampled uniformly
 430 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 431 distribution on $[0, 1]$.

432 We assume that prediction events occur on a substantially faster timescale than
 433 social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$
 434 of each player analytically rather than explicitly sampling prediction events. The
 435 expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the
 436 stochastic AI error ϵ , conditional on the current population strategies and the fixed
 437 target parameters. Closed form expression of expected payoffs are derived for all
 438 incentive structures and are presented in the Supplementary Methods.

439 Each generation consists of an expected payoff calculation followed by one op-
 440 portunity for pairwise social learning. Two distinct players, A and B , are randomly
 441 sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

442 where s represents the strength of social learning and was set to 50 in all simula-
 443 tions. We introduce a mutation rate μ , such that with probability μ , player A does
 444 not imitate player B but instead redrew all strategy components from the same dis-
 445 tributions used to initialize the population. The mutation rate is set to zero in all
 446 simulations except those presented in Supplementary Fig. S4.

447 Unless otherwise specified, each simulation is run for 200,000 generations. When a
 448 result is represented by a single summary value rather than as a temporal trajectory,
 449 the metric was averaged over the final 10,000 generations.

450 To evaluate the dynamics of collective prediction, we define six summary met-
 451 rics: collective accuracy, collective bias, collective variance, counterfactual collective
 452 accuracy, interest diversity, and median reliance on AI. At each population state,
 453 expectations and variances are evaluated analytically over the distributions of the
 454 causal factors and AI errors rather than estimated from sampled prediction events.
 455 Collective bias and collective variance constitute the bias–variance decomposition of
 456 the group level collective prediction error.

457 Collective accuracy is normalized relative to the baseline prediction α_0 . A nega-
 458 tive value indicates that the collective prediction performs worse than the baseline.
 459 Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the
 460 evolved interests and beliefs of the AI-assisted population. In calculating interest
 461 diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

462 **Data and code availability**

463 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
464 `parklab-snu/collective-intelligence-with-AI`).

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469 **Contributions**

470 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
471 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
472 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
473 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

474 **Competing Interests**

475 The authors declare no competing interests.

476 Supplementary Text

477 Here, we derive the expected payoffs and collective accuracy used in the simulations.
478 We consider Omniscient AI and Chatbot AI separately because they contain differ-
479 ent amounts of information and therefore require different aggregation rules. We
480 then analyze the corresponding evolutionary dynamics under a deterministic large-
481 population approximation.

482 S1 Omniscient AI

483 We first consider Omniscient AI, which incorporates all causal factors regardless of
484 the player's interest. We derive individual and collective predictions before calculat-
485 ing the expected payoffs under each incentive structure.

486 S1.1 Model

487 Because Omniscient AI observes all M causal factors, its prediction for player A is
488 given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (\text{S1})$$

489 where b_i represent factor-specific biases and ϵ represents stochastic error, following
490 a Gaussian distribution with mean zero and standard deviation τ . A new stochastic
491 error is generated independently for each player and prediction event.

492 Let $B = \sum_{i=0}^M b_i$ denote the total additive bias. Because the true prediction
493 target is $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (\text{S2})$$

494 In contrast to the AI prediction, player A 's initial prediction depends only on the
495 causal factor associated with their interest. Their initial and revised predictions are
496 therefore given by:

$$Y_A = c_A X_{e_A} \quad (\text{S3})$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (\text{S4})$$

497 represents player A 's interest, c_A represents their belief about the effect of the corre-
498 sponding factor, and β_A represents their reliance on AI. As in the main text, we set
499 $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

500 S1.2 Aggregation

501 Because the AI component of each revised prediction contains information about all
502 M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (S5)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (S6)$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j}\} \quad (S7)$$

503 To separate the contributions of AI and the players' initial predictions, we define the
504 population's mean reliance on AI, the stochastic AI error remaining after aggregation,
505 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S8)$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (S9)$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (S10)$$

506 Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error
507 remaining after averaging, and γ_i represents the contribution of players' initial pre-
508 dictions to factor i . Using these definitions, the revised collective prediction can be
509 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (S11)$$

510 Thus, the collective prediction consists of an AI-derived component, the contributions
511 of the players' initial predictions, and the stochastic AI error that remains after
512 aggregation.

513 S1.3 Expected payoffs

514 We next derive the expected payoff under each incentive structure. Expectations are
515 taken over the distributions of the causal factors and stochastic AI errors, conditional
516 on the current population strategies and the fixed parameters of the prediction target.

517 Feedback

518 Under the feedback structure, a player's payoff depends on the alignment between
 519 their revised prediction and the residual error of the collective prediction. To express
 520 this residual, we first define the remaining error in the intercept and in the coefficient
 521 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S12})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S13})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S14})$$

522 where δ_0 represents the remaining error in the intercept, whereas δ_i represents the
 523 remaining error associated with factor i . Because the causal factors and stochastic AI
 524 errors are independent and have zero mean, cross terms vanish when the expectation
 525 is taken. Substituting the revised individual prediction and collective residual into
 526 the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S15})$$

$$= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S16})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^m \delta_i X_i - \bar{\eta} \right\} \quad (\text{S17})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S18})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S19})$$

527 The first term represents the alignment between the AI-derived prediction and the
 528 collective residual, whereas the second term represents the contribution of the player's
 529 initial prediction. The final term arises because the stochastic error in player A 's AI
 530 prediction also contributes to the collective prediction. This finite-population effect
 531 decreases as population size increases.

532 Niche-expert

533 Under the niche-expert structure, the payoff is the negative squared error of the
 534 revised individual prediction, weighted by the frequency of the player's interest.
 535 Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S20})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S21})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S22})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S23})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S24})$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A} . The expected payoff is therefore:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S25})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S26})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)\alpha_0 + \beta_A B$$

$$+ (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\}^2] \quad (\text{S28})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S29})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S31})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re})\right] + \lambda\beta_A, \quad (\text{S32})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2\right] + \lambda\beta_A. \quad (\text{S33})$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

549 S1.4 Expected accuracy

550 Using the residual defined above, the mean squared error of the revised collective
551 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S34})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S35})$$

552 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S36})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S37})$$

553 The collective prediction error consists of the squared error in the intercept, the re-
554 maining error associated with each causal factor, and the stochastic AI error remain-
555 ing after averaging across players. Counterfactual collective accuracy is obtained
556 from the same expression by setting $\beta_A = 0$ for all players while retaining their
557 evolved interests and beliefs.

558 S2 Chatbot AI

559 Unlike Omniscient AI, Chatbot AI use only the causal factor associated with each
560 player's interest. This causes the information contained in AI predictions to depend
561 directly on the distribution of interests in the population. Here, we derive the collec-
562 tive prediction, expected payoffs, and expected collective accuracy under this form
563 of AI assistance.

564 S2.1 Model

565 The prediction generated by Chatbot AI for player A depends on the causal factor
566 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S38})$$

567 where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error.
568 The stochastic error follows a Gaussian distribution with mean zero and standard
569 deviation τ and is generated independently across players and prediction events. The

570 player's initial and revised predictions are given by Player A's initial and revised
 571 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S39})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S40})$$

572 Here, c_A represents player A's belief about the effect of their focal factor, e_A repre-
 573 sents their interest, and β_A represents their reliance on AI. Thus, the revised predic-
 574 tion remains specific to the single factor associated with the player's interest.

575 S2.2 Aggregation

576 Because both the player's initial prediction and the Chatbot AI prediction contain
 577 information about the same causal factor, we use clustering aggregation. Revised
 578 predictions are first averaged among players interested in the same factor and then
 579 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S41})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S42})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon], \quad (\text{S43})$$

580 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of
 581 players in that set.

582 To simplify the collective prediction, we define the effective coefficient, weighted
 583 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S44})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S45})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S46})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon, \quad (\text{S47})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S48})$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S49})$$

where $\mathcal{I} = \{i \in \{0, \dots, M\} > 0\}$ denotes the set of represented interests.

S2.3 Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S50})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S51})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S52})$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S53})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S54})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S55})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S56})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

604 Niche-expert

605 Under the niche-expert structure, the negative squared error of each revised indi-
 606 vidual prediction is weighted by the frequency of the player's interest. We consider
 607 $e_A = 0$ separately because these players make constant predictions rather than factor-
 608 dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S57})$$

$$= -\mathbb{E} \left[\rho_0 \left[\left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S58})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0)\beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S59})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S60})$$

609 When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The
 610 expected payoff is then

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S61})$$

$$= -\mathbb{E} [\rho_{e_A} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} \quad (\text{S62})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S63})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S64})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S65})$$

611 In both cases, the expected squared error consists of the mismatch between the
 612 player's belief and the true coefficient, contributions from causal factors omitted
 613 from the individual prediction, factor-specific AI bias, and stochastic AI error. The
 614 niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's
 615 interest.

616 Incentivizing or penalizing AI use

617 Directly incentivizing or penalizing AI use adds the term $\lambda \beta_A$ to the player's payoff.
 618 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}(\pi_{A,Feedback}) = \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S66})$$

$$= \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S67})$$

$$\mathbb{E}(\pi_{A,Niche-expert}) = -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S68})$$

$$= -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S69})$$

Balanced

Under the balanced structure, the payoff corresponds to the arithmetic mean of the feedback and niche-expert payoff components:

$$\mathbb{E}(\pi_A) = \mathbb{E} \left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2} \right] \quad (\text{S70})$$

$$= \frac{\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - \mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right]}{2} \quad (\text{S71})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

S2.4 Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept α_0 serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S72})$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S73})$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative

635 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S74})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S75})$$

636 Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance
637 equal to the baseline, and $R < 0$ represents performance worse than the baseline.

638 S3 Evolutionary dynamics

639 Following previous work (Wang et al., 2025; Taylor and Jonker, 1978; Traulsen et al.,
640 2005), we use replicator dynamics as a deterministic large-population approximation
641 to characterize equilibrium states and their local stability. The agent-based sim-
642 ulations themselves follow the Fermi imitation rule defined above. At equal-payoff
643 stationary states, the replicator and Fermi formulations imply the same local invasion
644 directions for any finite selection strength $s > 0$, although their transient dynamics
645 need not coincide. We focus on the feedback and niche-expert structures, for which
646 the long-term dynamics can be characterized analytically. The direct interventions
647 on AI use and the balanced structure are examined through simulations.

648 Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff.
649 Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S76})$$

650 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$ represents the mean expected payoff of the population. At an
651 internal stationary state, all strategies with positive frequency must have the same
652 expected payoff. A boundary state can also be stationary because a strategy with zero
653 frequency cannot increase unless it is introduced into the population. We therefore
654 assess the stability of boundary states by asking whether they can be invaded by a
655 rare alternative strategy.

656 S3.1 Omniscient AI

657 With Omniscient AI, every AI prediction contains information about the full predic-
658 tion target, and revised predictions are combined through averaging. As population
659 size increases, independent AI errors vanish from the collective prediction. How-
660 ever, these errors remain in individual prediction errors and can therefore continue
661 to affect niche-expert payoffs. We consider the two incentive structures separately.

662 S3.1.1 Feedback

663 Under the feedback structure, both the stochastic error in the collective prediction,
 664 $\bar{\eta}$, and its contribution to expected payoff vanish as $N \rightarrow \infty$. Following previous
 665 work (Wang et al., 2025), we define the Lyapunov function as the expected error of
 666 collective prediction:

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S77})$$

667 The function V is non-negative and equals zero when the collective prediction is
 668 identical to the target. To determine how V changes over time, we take its time
 669 derivative:

$$\frac{dV}{dt} = -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \quad (\text{S78})$$

$$= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \quad (\text{S79})$$

$$= -2 \left[\left\{ (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right]. \quad (\text{S80})$$

670 For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S81})$$

671 Although K changes with the population state, it is common to all strategies at a
 672 given state and can therefore be treated as a constant with respect to the strategy
 673 index λ . We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S82})$$

674 Under the replicator equation, the rate of change in the population mean of a
 675 strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S83})$$

676 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta)cG_e, u). \quad (\text{S84})$$

677 Therefore,

$$\frac{dV}{dt} = -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta)cG_e, u) \} \quad (\text{S85})$$

$$= -2 \text{Cov}(K\beta + (1 - \beta)cG_e, u). \quad (\text{S86})$$

For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy λ is

$$u_\lambda = K\beta_\lambda + (1 - \beta_\lambda)c_\lambda G_{e_\lambda}. \quad (\text{S87})$$

This expected payoff is identical to the quantity inside the covariance above. Therefore,

$$\frac{dV}{dt} = -2\text{Cov}(u, u) = -2\text{Var}(u) \leq 0. \quad (\text{S88})$$

Because payoff variance is non-negative, the expected collective prediction error cannot increase and decreases whenever the strategies currently represented in the population differ in their expected payoffs. The derivative becomes zero when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition arises when all represented strategies receive the same expected payoff. However, this condition does not necessarily imply that $V = 0$, because a population can lack strategies that would improve its collective prediction.

If $V > 0$, at least one component of the collective residual remains nonzero. A rare strategy whose prediction is aligned with this residual can obtain a higher expected payoff than the resident strategies and invade the population. Thus, provided that novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$. At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

S3.1.2 Niche-expert

Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each individual. Stochastic AI error therefore remains in the expected payoff even when $N \rightarrow \infty$, and we retain $\tau > 0$ throughout this analysis.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S89})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its

unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S90})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

To compare the optimized strategies across interests, let L_e denote the minimum expected individual squared error within interest e , $\mathbb{E}[(Y_A^{re} - Y)^2]$. For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S91})$$

If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating the expected squared prediction error at the corresponding boundary.

Interest 0 must be considered separately because $X_0 = 1$ and the player's belief determines a constant prediction. Optimizing with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S92})$$

Substituting this into the expected payoff gives a strictly concave function of β , with the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S93})$$

Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S94})$$

and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S95})$$

At this optimum, the player adjusts their belief to offset the systematic AI bias. The stochastic component of AI error cannot be removed in the same way and therefore limits the benefit of AI reliance.

We denote the optimal strategy within interest e by S_e . When $\beta_e^* = 1$, different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class. A stable state can contain only the optimal strategy class within each represented interest because a suboptimal strategy can be invaded by S_e .

737 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
738 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S96})$$

739 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator equa-
740 tion at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S97})$$

741 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S98})$$

742 This internal stationary state is locally stable. At equilibrium, all represented in-
743 terests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
744 interest increases slightly above its equilibrium value, its payoff decreases because
745 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of an
746 interest decreases below its equilibrium value, its payoff increases. The replicator dy-
747 namics therefore generate a restoring force that reduces small deviations in interest
748 frequencies and drives the population back toward the internal stationary state.

749 If one or more interests are absent, the resulting boundary state can be stationary
750 under the replicator equation, but it is unstable against invasion. The payoff of a rare
751 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
752 population is negative when $L_e > 0$, so the rare interest increases when introduced
753 at sufficiently small frequency.

754 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
755 prediction strategies and their frequencies. Whether the constrained optimum has
756 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
757 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
758 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
759 interest frequency remain uniquely determined.

760 Having characterized this equilibrium, we next compare its collective prediction
761 with a prediction made entirely by AI. In particular, the expected squared error of
762 an Omniscient AI prediction is

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S99})$$

763 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
764 fore, because L_e is the minimum individual expected squared prediction error within
765 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S100})$$

Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S101})$$

Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S102})$$

The collective prediction is the average of individual revised predictions. By convexity of the squared error,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S103})$$

Therefore,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] < \mathbb{E} \left[(Y - Y^{\text{AI}})^2 \right]. \quad (\text{S104})$$

Thus, the diversity of optimized predictions allows the population to outperform an individual AI prediction at equilibrium.

However, outperforming AI does not imply perfect collective accuracy. Under averaging, the non-AI contribution associated with a particular factor is provided only by individuals interested in that factor. The predictions of individuals with other interests therefore attenuate that factor's coefficient in the collective prediction. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \left\{ 1 - \bar{\beta}^* - \rho_i^* (1 - \beta_i^*) \right\} = \alpha_i \sum_{e \neq i} \rho_e^* (1 - \beta_e^*). \quad (\text{S105})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S106})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S107})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$. Thus, the niche-expert structure preserves interest diversity and allows the population to outperform AI, but averaging prevents the population from reaching perfect collective accuracy.

Supplementary Figure

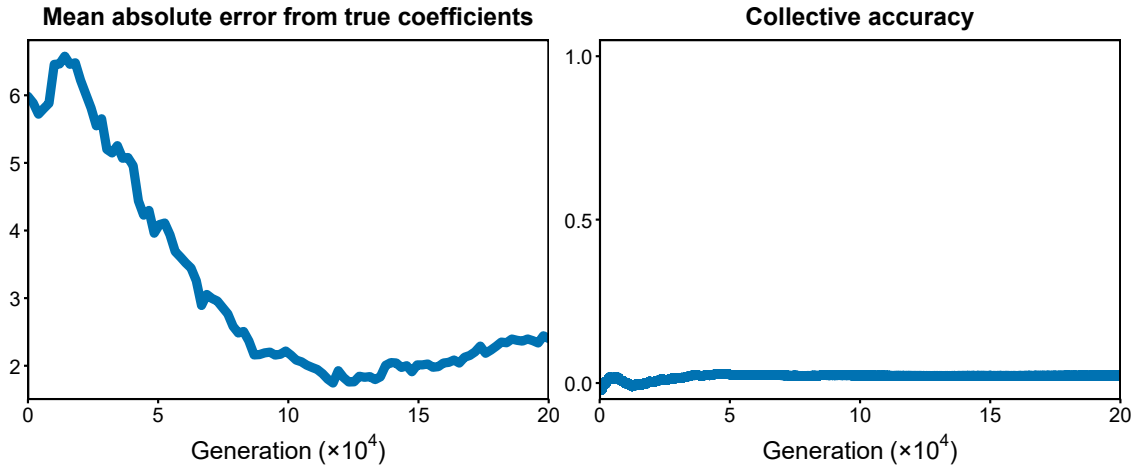


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $m + 1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$.

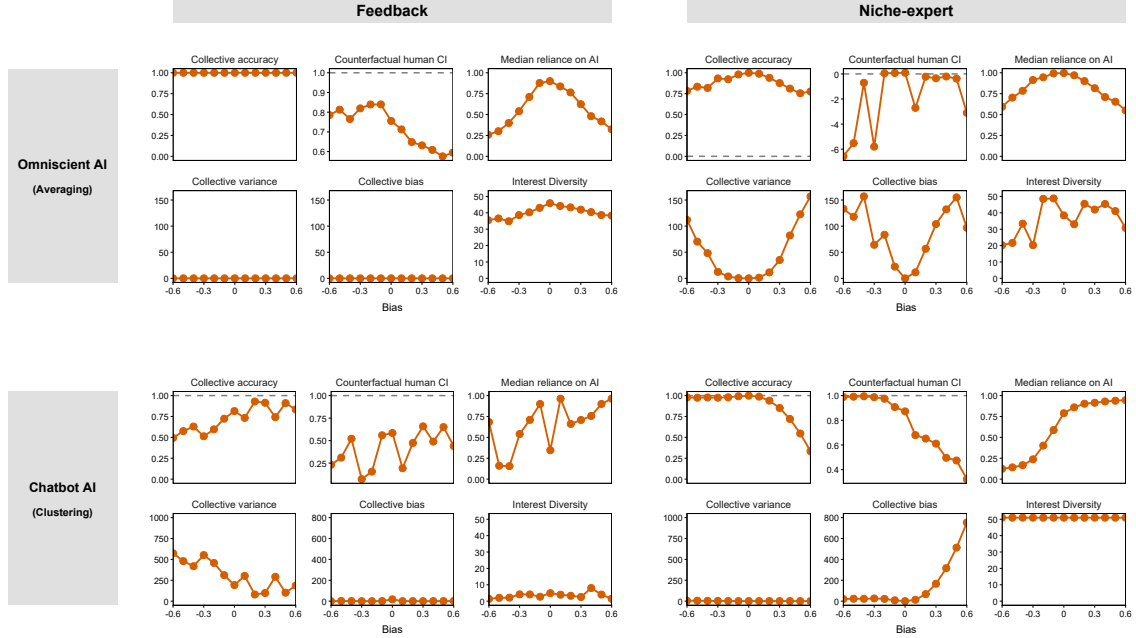


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (Omniscient AI, and Chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

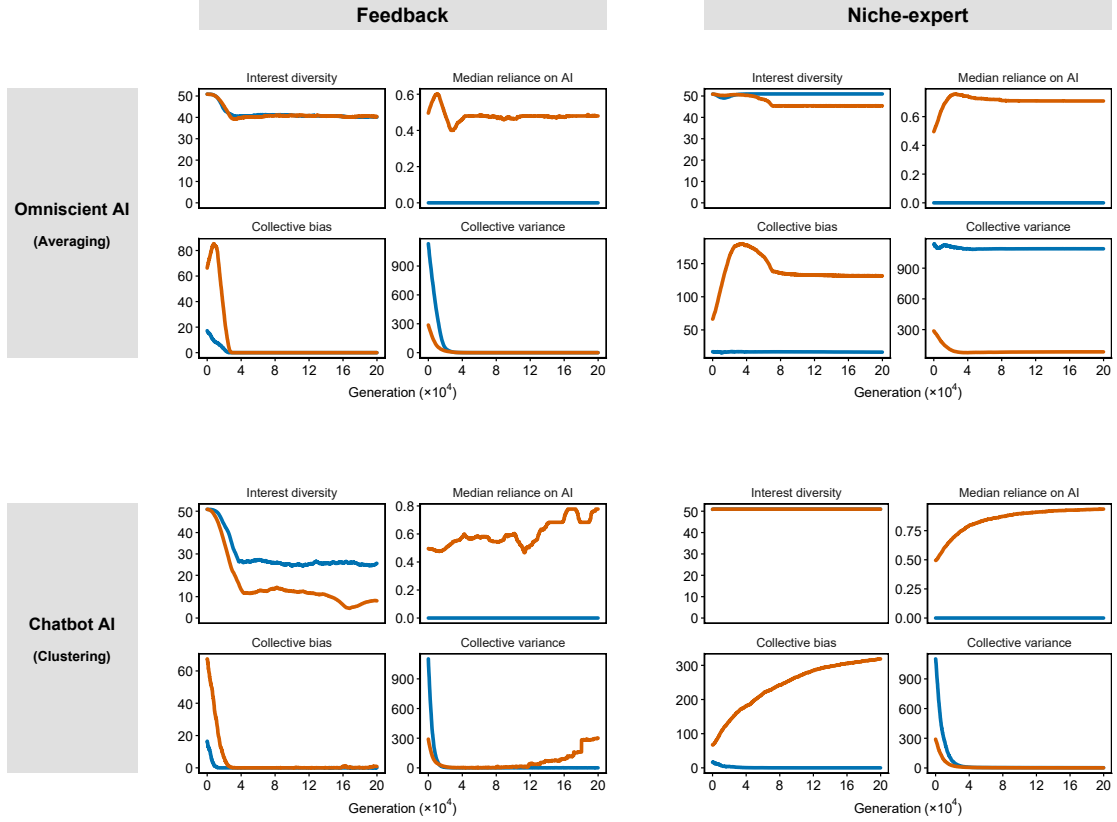


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as Hill-Shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance represent variability of the collective prediction error across prediction events. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

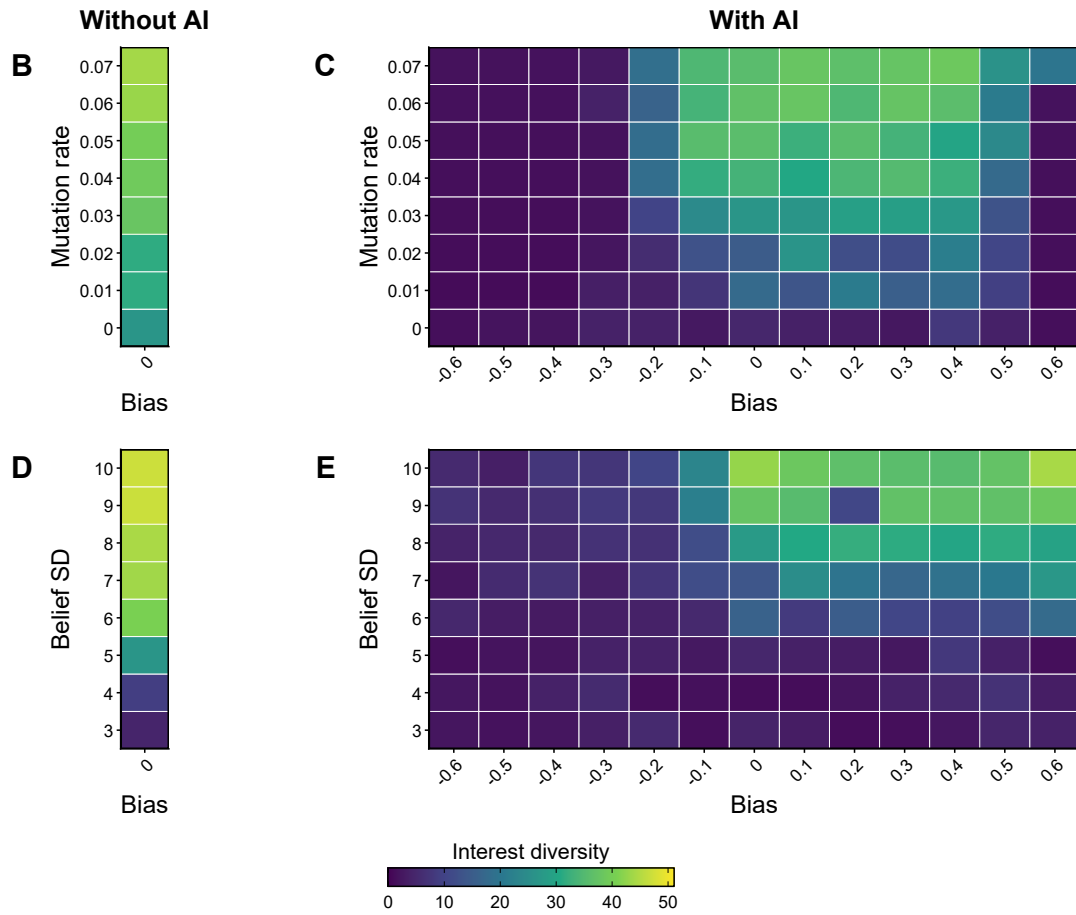
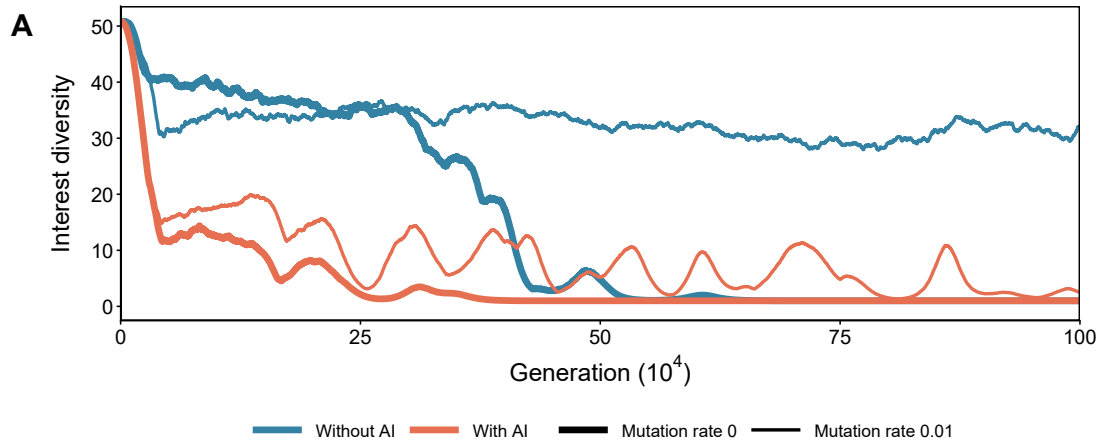


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, Chatbot AI is used.

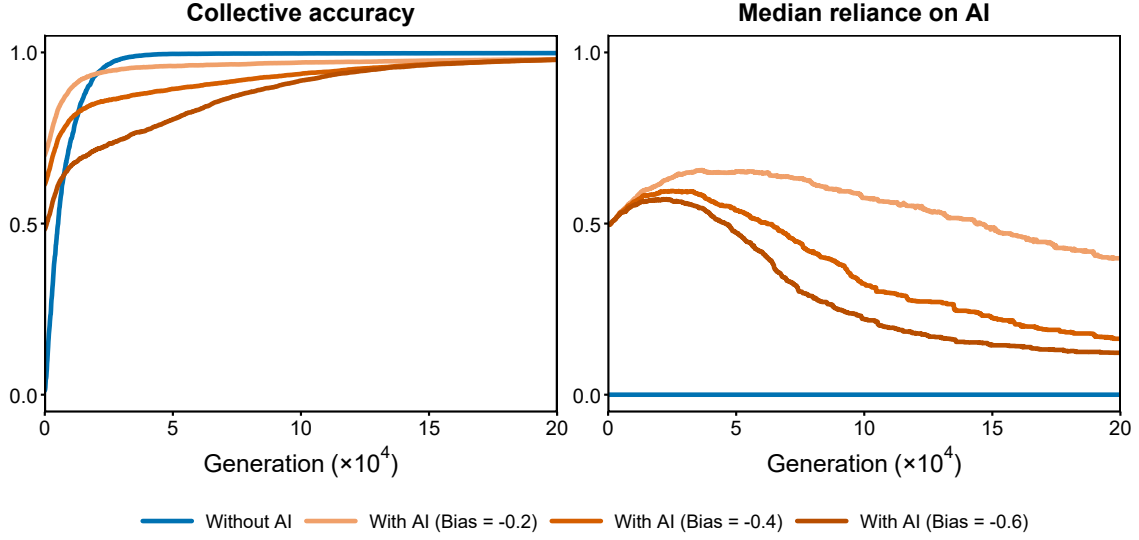


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

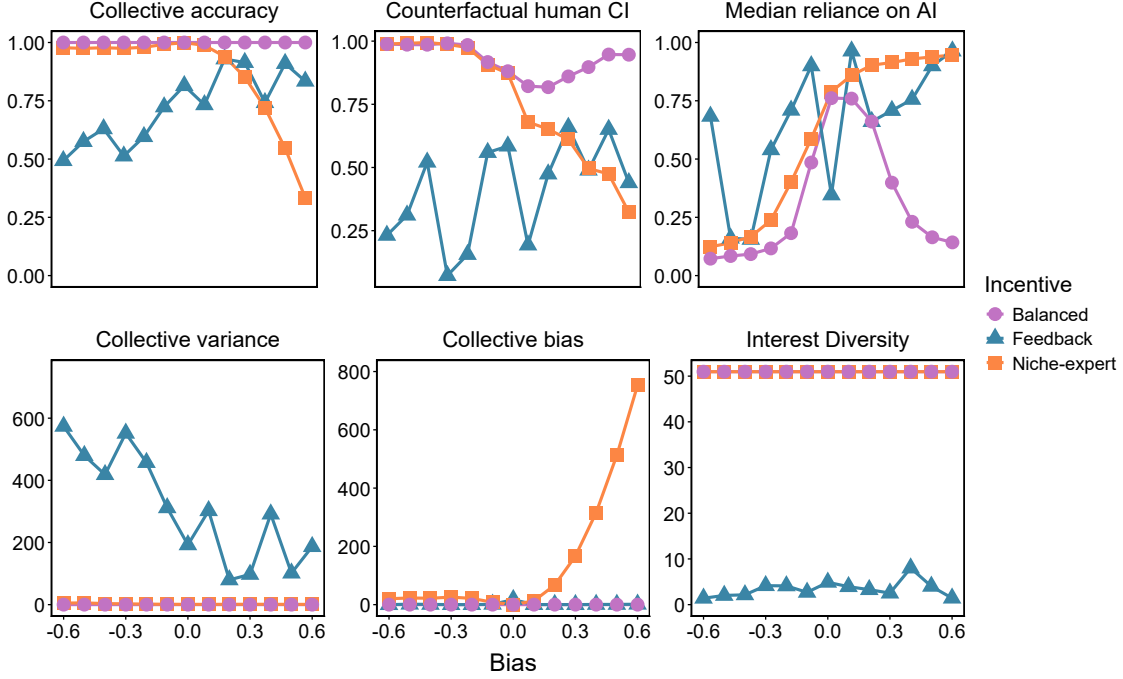


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,000–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

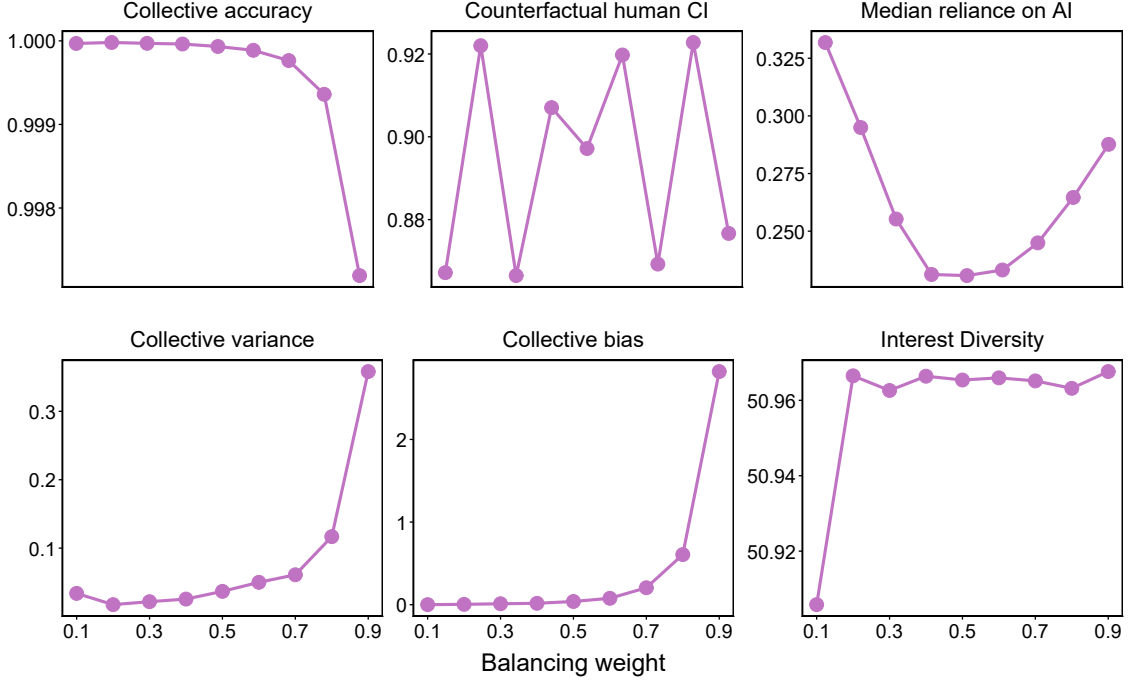


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, Chatbot AI is used.

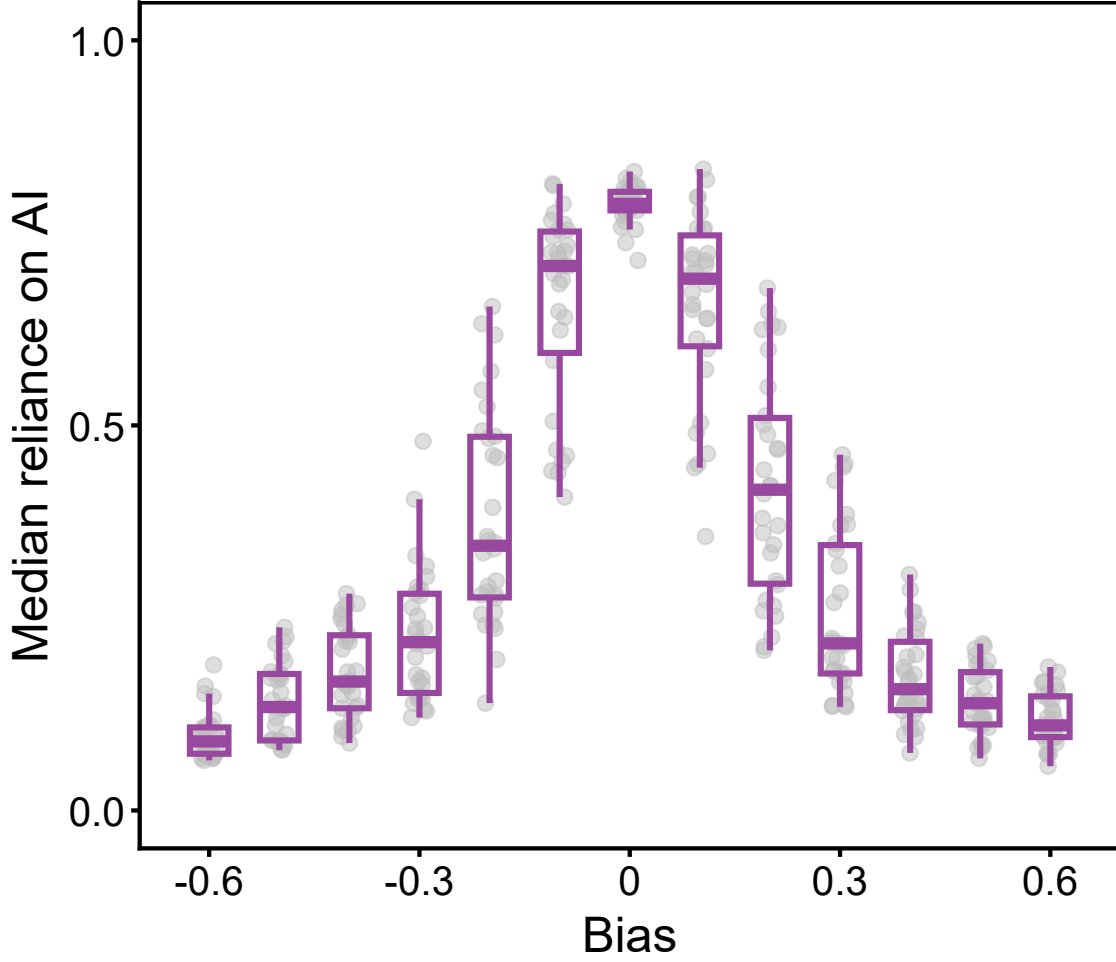


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

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